

Submission C solution to Problem 8

Abstract

We prove that if the lattice of flats of a finite matroid M admits an order-reversing involution $F \mapsto F^\perp$, then the relative Dressian

$$\mathrm{Dr}(M) = \{\text{tropical linear subspaces of } \mathrm{Trop}(M)\}$$

admits an order-reversing involution extending $F \mapsto F^\perp$ under the natural embedding

$$\mathcal{L}(M)^{\mathrm{op}} \hookrightarrow \mathrm{Dr}(M), \quad F \longmapsto \mathrm{Trop}(M/F \oplus U_{0,F}).$$

The main point is that an order-reversing involution of a geometric lattice forces the lattice to be modular. For modular matroids, valuated quotients are naturally valuated matroids on the modular lattice of flats, and the desired involution is obtained by pulling back along the lattice polarity.

1 Conventions and background

Let M be a finite matroid of rank d on ground set E . We write

$$\mathcal{L}(M)$$

for its lattice of flats, ordered by inclusion. If $F \in \mathcal{L}(M)$, then

$$X_F := \mathrm{Trop}(M/F \oplus U_{0,F})$$

is a tropical linear subspace of $\mathrm{Trop}(M)$. The assignment $F \mapsto X_F$ is order-reversing, hence gives an embedding

$$\mathcal{L}(M)^{\mathrm{op}} \hookrightarrow \mathrm{Dr}(M).$$

We use the valuated-matroid model of tropical linear spaces. A valuated matroid p of rank s on E determines a tropical linear space $\mathrm{Trop}(p)$ of dimension $s - 1$, and tropical linear spaces are identified up to adding a global scalar to p . If q and p are valuated matroids of ranks $t \geq s$, then

$$\mathrm{Trop}(p) \subseteq \mathrm{Trop}(q)$$

if and only if p is a valuated quotient of q , equivalently (q, p) satisfies the tropical incidence–Plucker relations. This is the valuated flag matroid correspondence; see Speyer [7], MacLagan–Sturmfels [4, Ch. 4], and Brandt–Eur–Zhang [2].

For a flat F , the corresponding valuated quotient p_F of rank $d - \mathrm{rk}(F)$ is the trivial valuation on the matroid $M/F \oplus U_{0,F}$. Thus, if $B \subseteq E$ has size $d - \mathrm{rk}(F)$, then

$$p_F(B) = \begin{cases} 0, & \text{if } B \text{ is a basis of } M/F, \\ \infty, & \text{otherwise.} \end{cases}$$

Equivalently, for an independent set B of M ,

$$p_F(B) < \infty \iff \mathrm{cl}_M(B) \vee F = E.$$

2 A lattice-theoretic consequence of polarity

We first record the key structural consequence of the existence of an order-reversing involution on the lattice of flats.

Lemma 2.1. *Let L be a finite ranked semimodular lattice of rank d . Suppose that L admits an order-reversing involution*

$$\alpha : L \longrightarrow L, \quad x \longmapsto x^\perp.$$

Then L is modular.

Proof. Since α is an order-reversing involution, it sends maximal chains to maximal chains with the order reversed. Hence

$$\text{rk}(x^\perp) = d - \text{rk}(x)$$

for all $x \in L$.

Because L is semimodular, for all $x, y \in L$ we have

$$\text{rk}(x) + \text{rk}(y) \geq \text{rk}(x \wedge y) + \text{rk}(x \vee y).$$

Apply semimodularity to $x^\perp$ and $y^\perp$. Using

$$x^\perp \wedge y^\perp = (x \vee y)^\perp, \quad x^\perp \vee y^\perp = (x \wedge y)^\perp,$$

we obtain

$$\begin{aligned} \text{rk}(x^\perp) + \text{rk}(y^\perp) &\geq \text{rk}((x \vee y)^\perp) + \text{rk}((x \wedge y)^\perp) \\ (d - \text{rk}(x)) + (d - \text{rk}(y)) &\geq (d - \text{rk}(x \vee y)) + (d - \text{rk}(x \wedge y)). \end{aligned}$$

Cancelling $2d$ gives

$$\text{rk}(x) + \text{rk}(y) \leq \text{rk}(x \wedge y) + \text{rk}(x \vee y).$$

Together with semimodularity, equality holds:

$$\text{rk}(x) + \text{rk}(y) = \text{rk}(x \wedge y) + \text{rk}(x \vee y).$$

This is precisely the rank form of modularity. □

Corollary 2.2. *If M admits an order-reversing involution on $\mathcal{L}(M)$, then $\mathcal{L}(M)$ is modular.*

Proof. The lattice of flats of a matroid is a finite geometric lattice, hence finite, ranked, atomic, and semimodular. Apply Lemma 2.1. □

3 Valuated quotients of modular matroids

We now use the standard modular-lattice form of valuated matroids.

Let M be modular of rank d . For $0 \leq s \leq d$, write

$$\mathcal{L}_s(M) := \{F \in \mathcal{L}(M) : \text{rk}(F) = s\}.$$

Lemma 3.1. *Let p be a rank- s valuated quotient of the trivial valuated matroid of M . If $B, B' \subseteq E$ are bases of the same rank- s flat of M , then*

$$p(B) = p(B').$$

Hence p descends to a function

$$\bar{p} : \mathcal{L}_s(M) \longrightarrow \mathbb{T}, \quad \bar{p}(F) = p(B)$$

for any basis B of F .

Proof. It is enough to prove the claim when B and B' differ by one element. Let

$$B = S \cup \{i\}, \quad B' = S \cup \{j\},$$

where $|S| = s - 1$, and assume

$$\text{cl}_M(S \cup \{i\}) = \text{cl}_M(S \cup \{j\}) =: F.$$

Extend $S \cup \{i\}$, equivalently $S \cup \{j\}$, to a basis of M by choosing a set C of size $d - s$ such that

$$S \cup \{i\} \cup C \quad \text{and} \quad S \cup \{j\} \cup C$$

are both bases of M .

Let

$$J := S \cup \{i, j\} \cup C.$$

Then $|J| = d + 1$. Consider the tropical incidence relation for the valuated quotient p , indexed by S and J :

$$\min_{e \in J \setminus S} \{p(S \cup \{e\}) + \mu(J \setminus \{e\})\}$$

must attain its minimum at least twice, where μ is the trivial valuation of M . For $e = i$ and $e = j$, the terms are respectively

$$p(S \cup \{i\}) + \mu(S \cup \{j\} \cup C) = p(B),$$

and

$$p(S \cup \{j\}) + \mu(S \cup \{i\} \cup C) = p(B').$$

For $e \in C$, the set

$$J \setminus \{e\} = S \cup \{i, j\} \cup (C \setminus \{e\})$$

is dependent in M , because i and j are dependent over S . Hence

$$\mu(J \setminus \{e\}) = \infty.$$

Thus the incidence relation reduces to saying that the minimum of the two values $p(B)$ and $p(B')$ is attained twice. Therefore $p(B) = p(B')$.

Since the basis graph of a matroid restriction $M|F$ is connected, any two bases of F are connected by such single-element exchanges. The result follows. $\square$

We will use the following standard form of valuated matroid duality on modular geometric lattices. It is the modular-lattice version of ordinary valuated-matroid duality. In the Boolean case it is exactly the usual operation

$$p^*(A) = p(E \setminus A).$$

For modular lattices, the same construction is obtained by replacing set-complement by the lattice anti-isomorphism. See Dress–Wenzel [3], Murota [5, Ch. 4], and the flag version in Brandt–Eur–Zhang [2].

Theorem 3.2 (Modular-lattice valuated duality). *Let M be a modular matroid of rank d , and let*

$$\alpha : \mathcal{L}(M) \longrightarrow \mathcal{L}(M), \quad F \longmapsto F^\perp$$

be an order-reversing involution. Let p be a rank- s valuated quotient of M , and let $\bar{p} : \mathcal{L}_s(M) \rightarrow \mathbb{T}$ be the descended function from Lemma 3.1. Define a function

$$\bar{p}^\perp : \mathcal{L}_{d-s}(M) \longrightarrow \mathbb{T}$$

by

$$\bar{p}^\perp(G) := \bar{p}(G^\perp).$$

Then $\bar{p}^\perp$ is the flat-valued form of a rank- $(d-s)$ valuated quotient $p^\perp$ of M .

Moreover:

- (i) $(p^\perp)^\perp = p$ up to tropical scalar;
- (ii) if q and p are valuated quotients of M with $q \twoheadrightarrow p$, then

$$p^\perp \twoheadrightarrow q^\perp;$$

- (iii) the construction is compatible with tropical linear spaces:

$$\text{Trop}(p) \subseteq \text{Trop}(q) \iff \text{Trop}(q^\perp) \subseteq \text{Trop}(p^\perp).$$

Proof. We recall the argument. By Lemma 3.1, every valuated quotient p of a modular matroid M is equivalently a valuated matroid on the modular lattice $\mathcal{L}(M)$, with coordinates indexed by rank- s flats. The tropical Plucker relations are local relations in length-four intervals of $\mathcal{L}(M)$, and the tropical incidence relations for a pair of quotients are the corresponding local flag-Plucker relations. An order-reversing lattice isomorphism sends length-four intervals to length-four intervals and interchanges rank s with rank $d-s$. Therefore pulling the flat-valued Plucker coordinates back along $F \mapsto F^\perp$ preserves the tropical Plucker relations and reverses the flag incidence relations.

Thus $\bar{p}^\perp$ is again a valuated quotient of M , now of rank $d-s$. Since $\perp$ is an involution on $\mathcal{L}(M)$, applying the construction twice gives back $\bar{p}$. Finally, the valuated flag relation $q \twoheadrightarrow p$ is carried to $p^\perp \twoheadrightarrow q^\perp$, which is precisely the asserted reversal of containment of tropical linear spaces. $\square$

4 Construction of the involution

Let $L = \text{Trop}(p) \in \text{Dr}(M)$, where p is a valuated quotient of M of rank s . Define

$$\Phi(L) := \text{Trop}(p^\perp),$$

where $p^\perp$ is the rank- $(d-s)$ quotient supplied by Theorem 3.2.

Equivalently, if $A \subseteq E$ is independent of size $d-s$, then

$$p^\perp(A) = \bar{p}(\text{cl}_M(A)^\perp),$$

and if A is dependent in M , then $p^\perp(A) = \infty$.

Theorem 4.1. *Let M be a finite matroid whose lattice of flats admits an order-reversing involution*

$$F \longmapsto F^\perp.$$

Then the map

$$\Phi : \text{Dr}(M) \longrightarrow \text{Dr}(M), \quad \text{Trop}(p) \longmapsto \text{Trop}(p^\perp),$$

is an order-reversing involution. Moreover, for every flat $F \in \mathcal{L}(M)$,

$$\Phi(\text{Trop}(M/F \oplus U_{0,F})) = \text{Trop}(M/F^\perp \oplus U_{0,F^\perp}).$$

Thus Φ extends the given involution $F \mapsto F^\perp$ on $\mathcal{L}(M)$.

Proof. By Corollary 2.2, M is modular. Hence Theorem 3.2 applies.

First, Φ is well-defined because tropical linear spaces are identified with valuated matroids up to tropical scalar, and the construction $p \mapsto p^\perp$ is unchanged by adding a global scalar to p .

Second, since $F \mapsto F^\perp$ is an involution on $\mathcal{L}(M)$, Theorem 3.2 gives

$$(p^\perp)^\perp = p$$

up to tropical scalar. Therefore

$$\Phi^2 = \text{id}_{\text{Dr}(M)}.$$

Third, suppose

$$\text{Trop}(p) \subseteq \text{Trop}(q).$$

Equivalently, $q \twoheadrightarrow p$. By Theorem 3.2,

$$p^\perp \twoheadrightarrow q^\perp.$$

Therefore

$$\text{Trop}(q^\perp) \subseteq \text{Trop}(p^\perp),$$

which is precisely

$$\Phi(\text{Trop}(q)) \subseteq \Phi(\text{Trop}(p)).$$

Thus Φ is order-reversing.

It remains to check that Φ extends the given flat involution. Let $F \in \mathcal{L}(M)$, and put $f = \text{rk}(F)$. Let p_F be the trivial valuation of the quotient $M/F \oplus U_{0,F}$, so p_F has rank $d - f$. In flat-valued form,

$$\bar{p}_F(H) = \begin{cases} 0, & H \vee F = E, \\ \infty, & H \vee F \neq E, \end{cases} \quad H \in \mathcal{L}_{d-f}(M).$$

Let $G \in \mathcal{L}_f(M)$. Then

$$\bar{p}_F^\perp(G) = \bar{p}_F(G^\perp).$$

Thus

$$\bar{p}_F^\perp(G) = 0 \iff G^\perp \vee F = E.$$

Applying the order-reversing involution gives

$$G^\perp \vee F = E \iff G \wedge F^\perp = \hat{0}.$$

Because $\mathcal{L}(M)$ is modular and

$$\text{rk}(G) + \text{rk}(F^\perp) = f + (d - f) = d,$$

the condition $G \wedge F^\perp = \hat{0}$ is equivalent to

$$G \vee F^\perp = E.$$

Therefore

$$\bar{p}_F^\perp(G) = 0 \iff G \vee F^\perp = E.$$

But this is exactly the flat-valued trivial valuation of the quotient

$$M/F^\perp \oplus U_{0,F^\perp}.$$

Hence

$$p_F^\perp = p_{F^\perp},$$

and consequently

$$\Phi(X_F) = X_{F^\perp}.$$

This proves the claimed extension property. □

Corollary 4.2. *The answer to the question is affirmative: every order-reversing involution on $\mathcal{L}(M)$ extends to an order-reversing involution on $\text{Dr}(M)$.*

References

- [1] F. Ardila and C. J. Klivans, *The Bergman complex of a matroid and phylogenetic trees*, J. Combin. Theory Ser. B **96** (2006), 38–49.
- [2] M. Brandt, C. Eur, and L. Zhang, *Tropical flag varieties*, Adv. Math. **384** (2021), 107695.
- [3] A. Dress and W. Wenzel, *Valuated matroids*, Adv. Math. **93** (1992), 214–250.
- [4] D. Maclagan and B. Sturmfels, *Introduction to Tropical Geometry*, Graduate Studies in Mathematics, vol. 161, American Mathematical Society, 2015.
- [5] K. Murota, *Discrete Convex Analysis*, SIAM Monographs on Discrete Mathematics and Applications, SIAM, 2003.
- [6] J. Oxley, *Matroid Theory*, 2nd ed., Oxford Graduate Texts in Mathematics, Oxford University Press, 2011.
- [7] D. Speyer, *Tropical linear spaces*, SIAM J. Discrete Math. **22** (2008), 1527–1558.